

Advance Project Title: Supply Chain Management Dashboard

Objective:

To monitor and analyze the full supply chain pipeline:

- ❖ Procurement trends
- ❖ Inventory health
- ❖ Supplier performance
- ❖ Shipment and delivery tracking
- ❖ Cost control & ☐ lead time optimization

Step 1: Understand and Prepare the Dataset

Assume you have an Excel file (Supply_Chain_Data.xlsx) with the following 4 sheets:

A. Purchase_Orders

| PO_ID | Date | Supplier_ID | Item_ID | Quantity | Unit_Cost | Total_Cost | Delivery_Date | Delivery_Status |

Tracks purchase orders placed by the company

B. Inventory

| Item_ID | Item_Name | Category | Stock_On_Hand | Reorder_Level | Unit_Price |

Shows current stock levels and reorder thresholds

C. Suppliers

| Supplier_ID | Supplier_Name | Region | Rating (1-5) |

Basic supplier information and quality rating

D. Shipments

| PO_ID | Shipment_Date | Actual_Delivery_Date | Carrier | Status |

Shipment tracking and delivery status

Step 2: Load Data into Power BI

1. Open Power BI Desktop
2. Go to **Home** → **Get Data** → **Excel**
3. Browse and select Supply_Chain_Data.xlsx

4. Choose all 4 sheets:

- ❖ Purchase_Orders
- ❖ Inventory
- ❖ Suppliers
- ❖ Shipments

5. Click **Load**

Step 3: Data Cleaning & Transformation (Power Query)

1. Click **Transform Data** to open Power Query Editor

For Each Table:

Ensure proper Data Types:

- ❖ Date → Date format
- ❖ Quantity, Cost, Rating → Decimal/Whole Number
- ❖ PO_ID, Item_ID → Text

Create Calculated Fields:

- ❖ **Total_Cost** (if missing): = Quantity * Unit_Cost
- ❖ **Lead_Time** in Shipments: = Actual_Delivery_Date - Shipment_Date

Fix inconsistent values:

- Standardize Delivery_Status and Shipment Status to be all lowercase or title case
- Rename columns (e.g., Qty → Quantity)

Remove blank/null rows and irrelevant columns

2. Click **Close & Apply**

Step 4: Define Relationships (Model View)

Go to **Model View** in the left panel and:

1. **Drag these relationships:**

- ❖ Purchase_Orders.Supplier_ID → Suppliers.Supplier_ID
- ❖ Purchase_Orders.Item_ID → Inventory.Item_ID
- ❖ Purchase_Orders.PO_ID → Shipments.PO_ID

2. **Confirm:**

- ❖ Cardinality = **1:Many**
- ❖ Cross filter direction = **Single**

This allows seamless linking between tables in visuals.

Step 5: Create Visualizations (Report View)

A. KPI Cards (Top Section)

Use **Card Visual** for:

- ❖ Total Purchase Orders = COUNT(Purchase_Orders[PO_ID])
- ❖ Total Procurement Cost = SUM(Purchase_Orders[Total_Cost])
- ❖ Average Lead Time = Use AVERAGE(Shipments[Lead_Time])
- ❖ On-Time Delivery Rate = Create DAX (shown below)
- ❖ Stock-Out Items = DAX filter (below reorder level)

B. Bar/Column Charts

Use **Stacked/Clustered Column Charts**:

- ❖ Procurement Cost by Supplier → Axis: Supplier_Name, Values: SUM(Total_Cost)
- ❖ Number of POs per Region → Axis: Region, Values: COUNT(PO_ID)
- ❖ Inventory Stock vs Reorder Level → Axis: Item_Name, Values: Stock_On_Hand vs Reorder_Level

C. Line Charts

- ❖ Monthly Procurement Trend → Axis: Month (Date), Values: SUM(Total_Cost)
- ❖ Average Lead Time Trend → Axis: Month, Values: Average(Lead_Time)

D. Stacked Bar Chart

- ❖ Quantity Ordered vs Delivered per Item → Item on X-axis, Bars for Ordered vs Delivered

E. Pie/Donut Charts

- ❖ Delivery Status (Delivered, Delayed, In Transit) → Count of each Status in Shipments
- ❖ Supplier Distribution by Region → Count of Supplier_ID grouped by Region

F. Tables or Matrix

- ❖ Item-wise Inventory and Reorder Details → Matrix with Item_Name, Stock_On_Hand, Reorder_Level

❖ Supplier PO Summary → Table of Supplier_Name, #POs, Total_Cost, Avg_Rating

Step 6: Add Filters and Slicers

Use **Slicers** to dynamically filter visuals by:

- Supplier Name
- Region
- Category
- Item Name
- Delivery Status
- Month/Year (use Date Hierarchy)

Step 7: Create Key DAX Measures

Total Procurement Cost = SUM(Purchase_Orders[Total_Cost])

Average Lead Time =
AVERAGE(Shipments[Actual_Delivery_Date] - Shipments[Shipment_Date])

On-Time Delivery Rate (%) =
DIVIDE(
COUNTROWS(FILTER(Shipments, Shipments[Actual_Delivery_Date] <= Purchase_Orders[Delivery_Date])),
COUNTROWS(Shipments)
) * 100

Stock-Out Items =
COUNTROWS(FILTER(Inventory, Inventory[Stock_On_Hand] <= Inventory[Reorder_Level]))

Use **New Measure** in the Modeling tab to add these.

Step 8: Design the Dashboard Layout

Suggested Layout:

Section	Contents
Top	5 KPI Cards
Middle	Charts: Procurement, Inventory, Shipments
Bottom	Tables: Supplier Summary, Inventory Matrix

Tips:

- Use **icons**: (inventory), (delivery), (supplier)
- Stick to 2–3 **corporate colors** (e.g., blue, gray, green)
- Add titles and headers using **Text Box**

Step 9: Publish & Share

1. Save your dashboard as .pbix
2. Go to **File** → **Publish** → **My Workspace**
3. Login with Power BI account
4. From **Power BI Service**:
 - ❖ Create dashboards
 - ❖ Share with team members
5. Set **scheduled refresh** (if Excel is updated regularly)

Step 10: Add Advanced Enhancements

Drill-through:

Right-click on a supplier → View detail page with performance, lead time, issues

Bookmarks:

Create bookmark buttons for:

- ❖ Inventory View
- ❖ Supplier View
- ❖ Shipment View

Alert Indicators (Conditional Formatting):

Highlight:

- ❖ Items below reorder level
- ❖ Deliveries past due
- ❖ Low-rated suppliers (Rating < 3)

Final Output: Summary

Section	Insights
Procurement	Total spend, PO trends, cost by supplier
Inventory	Stock vs reorder level, stock-outs, top categories
Suppliers	Supplier rating, region distribution, PO count
Shipments	On-time vs delayed, carrier analysis, lead times